Dynamic Mean–Variance Portfolio Optimization with a Time-Varying Equity Universe

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Motivation and Research Question

- Portfolio optimization is theoretically appealing, but in practice it often struggles to outperform simple benchmarks out of sample.
- A central difficulty is that estimated expected returns are noisy, covariance matrices are unstable, and realistic trading frictions can make the optimizer behave conservatively.
- This project asks whether a rolling long-only mean–variance framework, applied to monthly U.S. equity data with a dynamic stock universe, can generate a portfolio that meaningfully differs from equal weight and performs better out of sample.
- A second goal is diagnostic: if the optimized portfolio looks too similar to equal weight, is that because the code is wrong, the optimizer is not converging, or the economic signal is simply weak?

Question

- Can a rolling long-only mean–variance portfolio improve on equal weight out of sample?
- The key difficulty is practical rather than theoretical:
  - expected returns are noisy,
  - covariance estimates are unstable,
  - turnover control can make the solution overly conservative.
- The project therefore studies both performance and diagnosis:
  - does the optimized portfolio outperform,
  - and if it remains close to equal weight, is that due to weak signal or implementation?

Data

- CRSP-style monthly equity panel.
- After standard filters:
  - 300 monthly dates,
  - 971 distinct securities.
- Keep common shares on major U.S. exchanges.
- Drop observations with missing returns.
- Reshape to a date-by-stock return panel.

Dynamic Universe

- The investable universe is defined recursively over time.
- At month t, a stock is eligible only if it has at least 60 prior monthly return observations.
- The number of eligible stocks is capped at 100 each month.
- This avoids defining the universe using full-sample availability and helps reduce look-ahead and survivorship concerns.
- After the initial burn-in period, the eligible universe reaches the 100-stock cap for most of the sample.

Optimization Problem

$$\max_{w_t} \quad \mu_t^\top w_t - \gamma\, w_t^\top \Sigma_t w_t - \kappa \|w_t - w_{t-1}\|_1$$

- μ_t: estimated expected return vector
- Σ_t: estimated covariance matrix
- γ: risk aversion
- κ: turnover penalty

subject to

$$w_{i,t} \geq 0, \qquad \sum_{i=1}^{N_t} w_{i,t} = 1$$

Implementation

- Moments are estimated from a 24-month rolling window.
- The problem is solved in PyTorch.
- Portfolio weights are parameterized through a softmax map:

$$w_{i,t} = \frac{\exp(v_{i,t})}{\sum_{j=1}^{N_t} \exp(v_{j,t})}$$

- This enforces:
  - nonnegativity,
  - full investment.
- When the universe changes, previous weights are aligned to the current stock set by security identifier.

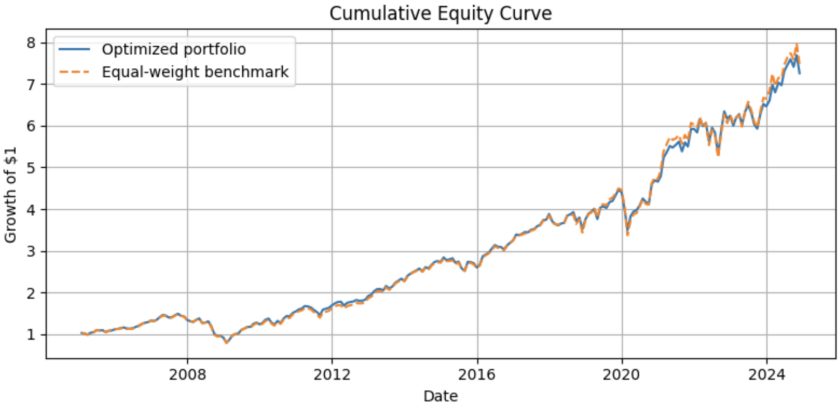
Backtest Design

- Rolling out-of-sample monthly backtest.
- At each date t:
  - identify the eligible stock universe,
  - estimate μ_t and Σ_t from the previous 24 months,
  - solve for w_t,
  - evaluate using realized returns in month $t+1$.
- Benchmark: equal-weight portfolio over the same date-specific universe.
- Total out-of-sample observations: 239 months.

Baseline Results

- Baseline hyperparameters:
  - $\gamma = 5.0$
  - $\kappa = 0.1$
  - learning rate $= 0.1$
  - steps $= 200$
- Optimized portfolio:
  - mean monthly return: 0.0092
  - annualized volatility: 0.1415
  - annualized Sharpe: 0.7774
  - cumulative return: 6.2591
- Equal-weight benchmark remains very close throughout.

Baseline Equity Curve



Baseline Equity Curve

- The optimized portfolio and equal-weight benchmark are nearly indistinguishable.
- The baseline specification remains close to an equal-weight regime.

Why So Close to Equal Weight?

- Three possible explanations:
  - the objective or constraints are implemented incorrectly,
  - the optimizer is not moving far from initialization,
  - the baseline specification is simply too conservative.
- In particular:
  - return estimates may be weak,
  - the risk penalty may dominate,
  - the turnover penalty may anchor the portfolio to past weights.
- This motivates a set of sanity checks.

Two-Asset Sanity Check

- Select one "good" stock and one "bad" stock from the full return panel.
- Criterion:
  - high versus low average realized monthly return,
  - at least one full rolling window of overlap.
- Selected pair:
  - good stock mean return: 0.0542
  - bad stock mean return: -0.0271
  - overlap: 34 months
- Goal: test whether the optimizer tilts toward the better asset rather than defaulting to 50–50.

Two-Asset Sanity Check Result

- Estimated mean returns over the last 24 overlapping months:

$$\mu_t = \begin{bmatrix} 0.0472 \\ -0.0022 \end{bmatrix}$$

- With $\kappa = 0$, the optimizer chooses:

$$w_t = \begin{bmatrix} 0.6089 \\ 0.3911 \end{bmatrix}$$

- This is already a clear deviation from equal weight.
- Conclusion: the optimizer is functioning correctly and is not mechanically producing 50–50 portfolios.

Convergence Check

- Relax penalties further:
  - $\gamma = 1.0$
  - $\kappa = 0$
- Increase optimization aggressiveness:
  - learning rate $= 2.0$
  - steps $= 2000$
- Result:

$$w_t = \begin{bmatrix} 0.9943 \\ 0.0057 \end{bmatrix}$$

- The optimizer can move far from equal weight when the signal is strong and the search is allowed to converge.

Implication of the Sanity Checks

- The baseline similarity to equal weight is not due to a coding error.
- The objective and constraints are implemented correctly.
- The optimizer can find strongly tilted portfolios in simple settings.
- Therefore, the baseline near-equal-weight behavior reflects:
  - conservative penalty choices,
  - limited optimizer movement,
  - weak cross-sectional signal in the full problem.

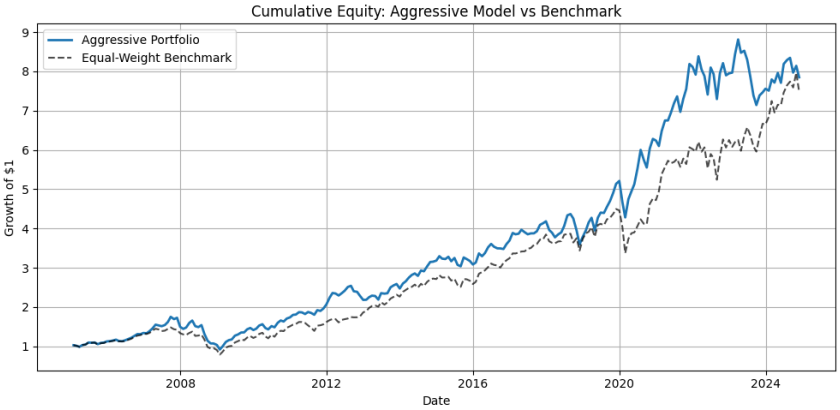
Aggressive Full-Sample Specification

- Re-run the full backtest with weaker turnover control.
- New specification:
  - $\gamma = 5.0$
  - $\kappa = 0.005$
  - learning rate $= 0.1$
  - steps $= 1000$
- Same data, same dynamic universe, same benchmark.
- Purpose: test whether the full strategy can move out of the near-equal-weight regime.

Aggressive Results

- Aggressive portfolio:
  - mean monthly return: 0.0096
  - annualized volatility: 0.1514
  - annualized Sharpe: 0.7626
  - cumulative return: 6.8492
- Equal-weight benchmark:
  - mean monthly return: 0.0094
  - annualized volatility: 0.1476
  - annualized Sharpe: 0.7620
  - cumulative return: 6.4788
- Higher cumulative return, but with slightly higher risk.

Aggressive Equity Curve



Aggressive Equity Curve

- The optimized portfolio now separates more clearly from equal weight.
- The gain is mainly in cumulative return rather than Sharpe ratio.

Portfolio Diagnostics

- Average L^1 distance from month-specific equal weight: 0.9049
- Average monthly turnover: 0.0289
- Under the baseline specification, turnover was only 0.0052.
- Hence the aggressive specification is genuinely more active:
  - larger deviations from equal weight,
  - more frequent reallocation through time.

Next Step: Hyperparameter Search

- The results are sensitive to the choice of hyperparameters.
- The next step is to search systematically over:
  - risk aversion γ,
  - turnover penalty κ,
  - learning rate,
  - number of optimization steps,
  - rolling window length.
- Objective:
  - identify which parameters drive deviation from equal weight,
  - and whether any specification improves out-of-sample performance robustly.